

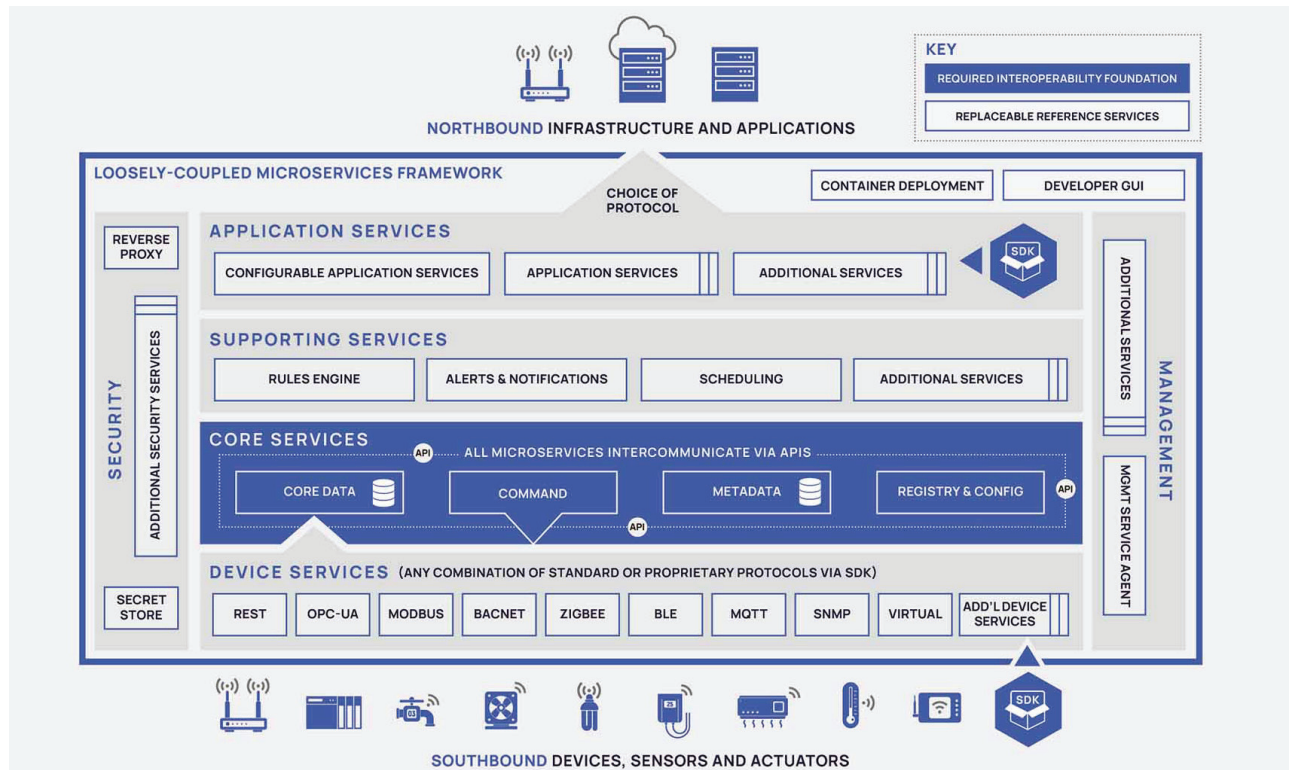


Figure 2: EdgeX focused IoT [Image credits: <https://www.edgexfoundry.org/cmsfiles/image/Platform/edgex-focused-at-the-iot-edge-2.jpg>]

resources, ensuring the efficient utilisation of assets. Ubuntu Core offers a range of features that bolster IoT deployments, including:

- **Transaction-based updates:** This enhances update reliability, crucial for uninterrupted IoT operations.
- **Secure boot and encryption:** These features fortify system security, ensuring data integrity and access control.
- **App store model:** Simplifying software management, it streamlines deployment and offers long-term support for mass-scale production.
- **Strict isolation with snaps:** Snaps, which are containerised, universal Linux packages, bundle applications and dependencies, introducing several benefits:
  - **Isolation:** Each snap operates in its own isolated environment, avoiding conflicts and enhancing security.
  - **Atomic updates:** Snaps can be individually updated, ensuring the latest security patches and bug fixes.
- **Dependency management:** Snaps encompass all necessary dependencies, simplifying software installation and maintenance.
- **Version rollback:** In case of issues with an update, snaps allow for straightforward rollback to a previous version.
- **Cross-distribution compatibility:** Snaps can be installed on any Linux distribution that supports them, promoting interoperability.
- **Long-term support for mass-scale production:** Organisations can rely on Ubuntu Core for an extended period, ensuring the stability and security of their IoT devices even in

large-scale, long-term production scenarios. This support typically includes regular updates, security patches, and maintenance services to keep the IoT infrastructure running smoothly and securely.

EdgeX Foundry, in combination with Ubuntu Core, is leading the charge in transforming IoT with edge computing. This powerful duo brings computing closer to the data source, enabling real-time processing and decision-making in a secure and efficient manner. As IoT continues to evolve, EdgeX Foundry's open source, flexible, and adaptable nature makes it the ideal choice for organisations looking to harness the full potential of edge computing in their IoT deployments. **END** 🐧

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The author has over 21 years of experience in the Linux and open source realm. He specialises in assisting clients in setting up private clouds using HCI based on ProxmoxVE and Ceph SDS.